

CS4423 - Networks

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3. Random Networks

Lecture 11: Properties of Random Graphs.

What are the typical features of a random network with n nodes and m edges?

- When does it contain a **tree** that's not a straight line?
- When does it contain a **triangle**, or indeed a **cycle** of any length? If so, how many?
- When does it contain a small **complete** graph of a given size?
- When does it contain a **large component**, larger than all other components?
- When does the network form a single **connected component**?
- How do these properties depend on n and m , or p ?

```
In [1]: import numpy as np
import pandas as pd
import networkx as nx
opts = { "with_labels": True, "node_color": 'y' }
```

Probability Distributions

- Denote by G_n the set of *all* graphs on the n points $X = \{0, \dots, n-1\}$.
- Regard the ER models A and B as **probability distributions** $P: G_n \rightarrow \mathbb{R}$.

Notation:

- $N = \binom{n}{2}$, the maximal number of edges of a graph $G \in G_n$.
- $m(G)$: the number of edges of a graph G .

* $G(n, m)$:

$$P(G) = \begin{cases} \binom{N}{m}^{-1}, & \text{if } m(G) = m, \\ 0, & \text{else.} \end{cases}$$

* $G(n, p)$:

$$P(G) = p^m (1 - p)^{N-m},$$

where $m = m(G)$.

Expected Size and Average Degree

In $G(n, m)$: * the expected **size** is

$$\bar{m} = m,$$

as every graph G in $G(n, m)$ has exactly m edges. * the expected **average degree** is

$$\langle k \rangle = \frac{2m}{n},$$

as every graph has average degree $2m/n$.

- Other properties of $G(n, m)$ are less straightforward, and it is easier to work with the $G(n, p)$.

In $G(n, p)$, with $N = \binom{n}{2}$: * the **expected size** is

$$\bar{m} = pN$$

* with **variance**

$$\sigma_m^2 = Np(1 - p);$$

* the expected **average degree** is

$$\langle k \rangle = p(n - 1).$$

* with **standard deviation**

$$\sigma_k = \sqrt{p(1 - p)(n - 1)}$$

- In particular, the **relative standard deviation** (or the **coefficient of variation**) of the size of a random model B graph is

$$\frac{\sigma_m}{\bar{m}} = \sqrt{\frac{1-p}{pN}} = \sqrt{\frac{2(1-p)}{pn(n-1)}} = \sqrt{\frac{2}{n\langle k \rangle} - \frac{2}{n(n-1)}},$$

a quantity that converges to 0 as $n \rightarrow \infty$ if $p(n-1) = \langle k \rangle$, the average node degree, is kept constant.

- In this sense, for large graphs, the fluctuations in the size of random graphs in model B can be neglected.
- Model B random graphs are used for theoretical purposes.
- Model A random graphs are preferred for sampling random graphs on a computer.
- A model A graph with n vertices and m edges corresponds to a model B graph with n vertices and $p = m/N$, where $N = \binom{n}{2}$.

Degree distribution

Definition. The **degree distribution** $p: \mathbb{N}_0 \rightarrow \mathbb{R}$, $k \mapsto p_k$ of a graph G is defined as

$$p_k = \frac{n_k}{n},$$

where, for $k \geq 0$, n_k is the number of nodes of degree k in G .

This definition can be extended to ensembles of graphs with n nodes (like the random graphs $G(n, m)$ and $G(n, p)$), by setting

$$p_k = \bar{n}_k/n,$$

where $\bar{n}_k$ denotes the expected value of the random variable n_k over the ensemble of graphs.

* The degree distribution in a random graph $G(n, p)$ is a **binomial distribution**:

$$p_k = \binom{n-1}{k} p^k (1-p)^{n-1-k} = \text{Bin}(n-1, p, k)$$

- Thus the **average degree** of a randomly chosen node is

$$\langle k \rangle = \sum_{k=0}^{n-1} k p_k = p(n-1)$$

with standard deviation

$$\sigma_k = \sqrt{p(1-p)(n-1)}$$

- When n is large, the typical p_k is a product of a binomial coefficient $\binom{n-1}{k}$ which can be very large, and the probabilities $p^k(1-p)^{n-1-k}$ which can be very small. This can lead to errors in the computation.
- In the limit $n \rightarrow \infty$, with $\bar{k} = p(n-1)$ kept constant, the binomial distribution $\text{Bin}(n-1, p, k)$ is well approximated by the **Poisson distribution**:

$$p_k = e^{-\lambda} \frac{\lambda^k}{k!} = \text{Pois}(\lambda, k),$$

where $\lambda = p(n-1)$.

```
In [2]: from math import factorial
        factorial(16)
```

```
Out[2]: 20922789888000
```

```
In [3]: # python 3.8 has:
        # from math import comb
```

$$\binom{n}{k} = \frac{n \cdot (n-1) \cdots (n-k+1)}{1 \cdot 2 \cdots k}$$

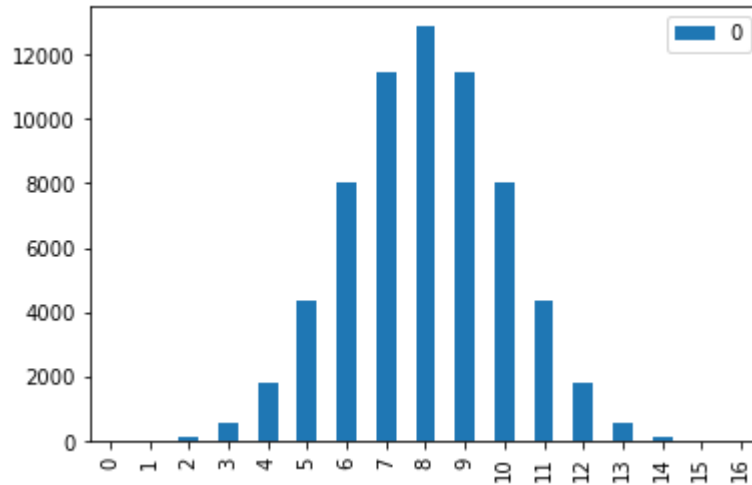
```
In [4]: def binomial(n, k):
        prod, top, bot = 1, n, 1
        for i in range(k):
            prod = (prod * top) // bot
            top, bot = top - 1, bot + 1
        return prod
```

```
In [5]: l = [binomial(16, k) for k in range(17)]
        print(l)
```

```
[1, 16, 120, 560, 1820, 4368, 8008, 11440, 12870, 11440, 8008, 4368,
1820, 560, 120, 16, 1]
```

```
In [6]: df = pd.DataFrame(l)
df.plot.bar()
```

Out[6]: <AxesSubplot:>



For n larger than k , [Stirlings formula](https://en.wikipedia.org/wiki/Stirling%27s_approximation) (https://en.wikipedia.org/wiki/Stirling%27s_approximation).

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

can be used to approximate a binomial coefficient as follows:

$$\binom{n}{k} = \frac{n \cdot (n-1) \dots (n-k+1)}{k!} \approx \frac{(n-k/2)^k}{k^k e^{-k} \sqrt{2\pi k}} = \frac{(n/k - 0.5)^k e^k}{\sqrt{2\pi k}}$$

```
In [7]: from math import exp, sqrt, pi, log
def binom_approx(n, k):
    return (n/k - 0.5)**k * exp(k) / sqrt(2 * pi * k)
```

```
In [8]: n = binomial(100, 2)
k = 50
print(binomial(n, k))
```

```
137831378168286642607524453056135424869947114595898731212427379458736
5832644708111045663868872218997025383324264236036878
```

```
In [9]: print(binom_approx(n, k))
print(binomial(n, k) / 10**120)
```

```
1.3739175898110523e+120
1.3783137816828663
```

Phase Transitions

- Point of view: for the random graph $G(n, p)$, suppose that $p = p(n)$ is a function of n , the number of nodes, and study the ensemble of graphs $G(n, p(n))$, as $n \rightarrow \infty$.
- Typically, $p(n) = cn^e$, for real numbers c, e .
- Then, to say that **almost every graph has a property Q** means that the probability of a graph in the ensemble to have property Q tends to 1, as $n \rightarrow \infty$.
- In other words, for any given property Q , there is a **critical probability function $p_Q(n)$** such that a.e. graph in $G(n, p(n))$ has property Q if $p(n)$ **decays slower** than $p_Q(n)$, and a.e. graph in $G(n, p(n))$ fails to have property Q if $p(n)$ **decays faster** than $p_Q(n)$.
- If Q is concerned with specific subgraphs, and if $p(n) \propto p_Q(n)$, we can determine the number of such subgraph contained in the network.
- This is made precise by the following theorem. (For a proof see Bollobás, *Random Graphs*, CUP 2001.)

****Theorem (Appearance of Subgraphs).**** Let F be a connected graph with a nodes and b edges. * If $p(n)/n^{-a/b} \rightarrow 0$ then almost every graph in the ensemble $G(n, p(n))$ does not contain a copy of F . * If $p(n)/n^{-a/b} \rightarrow \infty$ then almost every graph in the ensemble $G(n, p(n))$ does contain a copy of F . * If $p(n) = cn^{-a/b}$ then, as $n \rightarrow \infty$, the number n_F of F -subgraphs in G has distribution $\text{Pois}(\lambda, r)$, where $\lambda = c^b/|\text{Aut}(F)|$, with $|\text{Aut}(F)|$ being the number of *automorphisms* of F .

- So the critical probability function $p_Q(n)$ for Q being the appearance of a subgraph F with a nodes and b edges is $p_Q(n) = cn^{-a/b}$.

For example: * ****Trees**** with a nodes ($b = a - 1$) appear when $p(n) = cn^{-a/(a-1)}$. * ****Cycles**** of order a (where $b = a$) appear when $p(n) = cn^{-1}$. * ****Complete**** subgraphs of order a (where $b = \binom{a}{2} = \frac{1}{2}a(a-1)$) appear when $p(n) = cn^{-2/(a-1)}$.

- To determine n_F :
- there are $\binom{n}{a}$ ways to pick a nodes from G 's n nodes;
- between them, b links are present with probability p^b ;
- permuting the a nodes of F yields $a!/|\text{Aut}(F)|$ copies;
- in total, there are

$$n_F = \frac{n!p^b}{(n-a)!|\text{Aut}(F)|} \approx \frac{n^a p^b}{|\text{Aut}(F)|}$$

copies of F in G .

- Note how this number depends only on a and b .

Example: Numbers of * triads with $a = 3$ and $b = 2$: $3\binom{n}{3}p^2 = \frac{1}{2}n(n-1)(n-2)p^2 \approx \frac{1}{2}n^3 p^2$, * star graph with $a = 4$, $b = 3$ and $|\text{Aut}(F)| = 6$: $4\binom{n}{4}p^3 = \frac{1}{6}n(n-1)(n-2)(n-3)p^3 \approx \frac{1}{6}n^4 p^3$. * triangles where $a = b = 3$ and $|\text{Aut}(F)| = 6$: $\binom{n}{3}p^3 = \frac{1}{6}n(n-1)(n-2)p^3 \approx \frac{1}{6}n^3 p^3$.

- The actual **number of triads** in a graph G can be determined from its **degree distribution**, as each node of degree k is the central node of exactly $\binom{k}{2}$ triads (why?)
- `networkx` does not seem to have a function for that specific purpose, but `degree` is good enough.

```
In [10]: n, m = 100, 100
p = 2*m/n/(n-1)
print(f"Expect {n*(n-1)*(n-2) * p**2/2} triads")
```

Expect 197.979797979798 triads

```
In [11]: G = nx.gnm_random_graph(n, m)
sum(k*(k-1)//2 for k in dict(G.degree()).values())
```

Out[11]: 186

- The actual **number of triangles** in a graph G with adjacency matrix A is $\frac{1}{6}$ of the trace of A^3 (why?)
- Also, the `networkx` function `triangles` returns a dictionary, reporting for each node the number of triangles it is involved in.

```
In [12]: n, m = 100, 300
p = 2*m/n/(n-1)
print(f"Expect {n*(n-1)*(n-2) * p**3/6} triangles")
```

Expect 35.99632690541782 triangles

```
In [13]: G = nx.gnm_random_graph(n, m)
sum(nx.triangles(G).values())//3
```

Out[13]: 32

```
In [14]: A = nx.adjacency_matrix(G)
np.trace((A**3).toarray())//6
```

Out[14]: 32

* Moreover, $p(n) = \frac{1}{n} \ln n$ is the threshold probability for G to be connected. (This corresponds to $m = \frac{1}{2} n \ln n$ in model A .)

The Giant Connected Component

Definition (Giant Component). A connected component of a graph G is called a **giant component** if its number of nodes increases with the order n of G as some positive power of n .

Suppose $p(n) = cn^{-1}$ for some positive constant c . (Then the average degree $\langle k \rangle = pn = c$ remains fixed as $n \rightarrow \infty$.)

Theorem (Erdős-Rényi).

- If $c < 1$ the graph contains many small components, orders bounded by $O(\ln n)$.
- If $c = 1$ the graph has large components of order $O(n^{2/3})$.
- If $c > 1$ there is a unique **giant component** of order $O(n)$.

- In practice, in a given graph G of order n , a giant component appears when the average degree is 1, i.e., if $m = \frac{1}{2}n$, and then has order $n^{2/3}$ (e.g., $100^{2/3} = 21.5$).

Code Corner

networkx

- triangles : [\[doc\]](https://networkx.org/documentation/stable/reference/algorithms/generated/networkx.algorithms.cluster.triangles.html)
(<https://networkx.org/documentation/stable/reference/algorithms/generated/networkx.algorithms.cluster.triangles.html>)

Exercises

1. Design an experiment with random graphs of suitable degree and size to verify the predicted numbers of triads above.
2. Show that a node of degree k is the central node of exactly $\binom{k}{2}$ triads.
3. Design an experiment with random graphs of suitable degree and size to verify the predicted numbers of triangles above.
4. Show that the number n_{Δ} of triangles in an graph G with adjacency matrix A is $n_{\Delta} = \frac{1}{6} \text{tr}(A^3)$, where

$\text{tr}(B)$ = denotes the **trace** of a square matrix B , i.e., the sum of its diagonal entries.

$$\sum_i b_{ii}$$

5. Design an experiment with random graphs of suitable degree and size to verify the predicted numbers of star graphs on 4 nodes above.

In []: